

Reference-Based Alignment of Large Sequence Databases

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Motivation

- Defined over an alphabet Σ .
- Text documents.
- Biological sequences (DNA).
- Near homology search:
 - Deviation from Q does not exceed a threshold δ ($\delta \leq 15\%$).

Q: TCTAGGGCA



The diagram consists of seven vertical white lines. Each line starts at a letter in the query sequence 'TCTAGGGCA' and extends downwards to a corresponding letter in the target sequence '...ACTTAGCTGTAGTCGTCTATGGCATATGCATGCTGATCTCGTGCGTCATG...'. The lines connect the 'T' to 'T', 'C' to 'C', 'T' to 'T', 'A' to 'A', 'G' to 'G', 'G' to 'G', and 'C' to 'C'.

...ACTTAGCTGTAGTCGTCTATGGCATATGCATGCTGATCTCGTGCGTCATG...

String Matching

- Given:
 - S: collection of sequences defined over an alphabet Σ .
 - Q: query sequence defined over Σ .
 - D: similarity measure.
- Find the most similar subsequence in S.

Our focus: DNA

- S: a set of DNA sequences.
- Q: DNA sequence
 - with a small deviation from the database match.
 - within $\delta |Q|$, for $\delta \leq 15\%$.
 - can be large (up to 10,000 nucleotides).

The Edit Distance

■ Initialization:

		A	C	T	G
	0	1	2	3	4
A	1				
T	2				
C	3				

The Edit Distance

■ First column:

- Match = 0
- In/del/sub = 1

		A	C	T	G
	0	1	2	3	4
A	1	0			
T	2	1			
C	3	2			

The Edit Distance

- Second column:

		A	C	T	G
	0	1	2	3	4
A	1	0	1		
T	2	1	1		
C	3	2	1		

The Edit Distance

■ Final Matrix:

		A	C	T	G
	0	1	2	3	4
A	1	0	1	2	3
T	2	1	1	1	2
C	3	2	1	2	2

The Edit Distance

■ Alignment Path:

A = A - T C
B = A C T G

		A	C	T	G
	0	1	2	3	4
A	1	0	1	2	3
T	2	1	1	1	2
C	3	2	2	2	2

The Edit Distance: Subsequence matching

■ Initialization:

		A	C	T	G
	0	0	0	0	0
A	1				
T	2				
C	3				

The Edit Distance: Subsequence matching

■ Final Matrix:

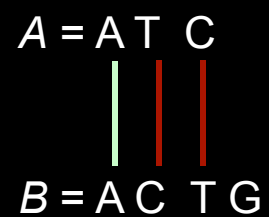
		A	C	T	G
	0	0	0	0	0
A	1	0	1	1	1
T	2	1	1	1	2
C	3	2	2	2	2

The Edit Distance: Subsequence matching

- One path:

		A	C	T	G
	0	0	0	0	0
A	1	0	1	1	1
T	2	1	1	1	2
C	3	2	1	2	2

A = A T C
B = A C T G



Smith-Waterman [Smith&Waterman et al. 1981]

- Is a similarity measure used for local alignment:
 - Match can be a subsequence of the query sequence.
- Define three penalties:
 - match, mismatch, gap.
 - Scoring parameters are defined by the user.
- Example:
 - $A = \text{ATC}$ and $B = \text{TATTCG}$
 - match = 2, mismatch = -1, gap = -1.

Smith-Waterman

■ Initialization:

		T	A	T	T	C	G
	0	0	0	0	0	0	0
A	0						
T	0						
C	0						
A	0						

Smith-Waterman

- First column:

		T	A	T	T	C	G
	0	0	0	0	0	0	0
A	0	-1					
T	0	2					
C	0	1					
A	0	0					

Smith-Waterman

- First column:

		T	A	T	T	C	G
	0	0	0	0	0	0	0
A	0	0					
T	0	2					
C	0	1					
A	0	0					

Smith-Waterman

- Second column:

		T	A	T	T	C	G
	0	0	0	0	0	0	0
A	0	0	2				
T	0	2	1				
C	0	1	1				
A	0	0	3				

The diagram illustrates the second column of a Smith-Waterman matrix. The matrix is a grid with rows and columns labeled with nucleotide bases (A, T, C, G). The values in the cells represent the score for a match or mismatch. The second column is highlighted, and green arrows indicate the path of maximum scores. The path starts at the cell (A, A) with a score of 2, moves down to (T, A) with a score of 1, then down to (C, A) with a score of 1, and finally down to (A, A) with a score of 3.

Smith-Waterman

■ Final Matrix:

		T	A	T	T	C	G
	0	0	0	0	0	0	0
A	0	0	2	1	0	0	0
T	0	2	1	2	3	2	1
C	0	1	1	1	2	5	4
A	0	0	3	2	1	4	4

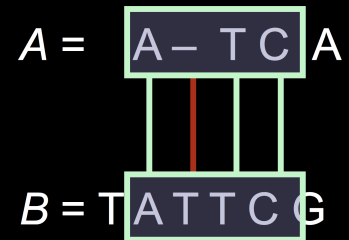
Smith-Waterman

- Detect highest value:

		T	A	T	T	C	G
	0	0	0	0	0	0	0
A	0	0	2	1	0	0	0
T	0	2	1	2	3	2	1
C	0	1	1	1	2	5	4
A	0	0	3	2	1	4	4

Smith-Waterman

■ Alignment Path:



		T	A	T	T	C	G
	0	0	0	0	0	0	0
A	0	0	2	1	0	0	0
T	0	2	1	2	3	2	1
C	0	1	1	1	2	5	4
A	0	0	3	2	1	4	4